MIL-F-18215(NOrd)
23 October 1954

MILITARY SPECIFICATION
FUSE, TIME, PLASTIC COATED

1. SCOPE

1.1 Scope. - This specification covers plastic coated, water-resistant, time fuse for use in pyrotechnics at temperatures ranging from -65°F to +160°F.

1.2 Classification. - Time fuse covered by this specification shall be of the following classes and types as specified in the contract or order.

Type I - Burning time 30 seconds per foot.
Type II - Burning time 45 seconds per foot.
Type III - Burning time 60 seconds per foot.

Class 1 - 30 grains of black powder per foot.
Class 2 - 40 grains of black powder per foot.

2. APPLICABLE DOCUMENTS

2.1 The following specifications, standards, drawings and publications of the issue in effect on the date of invitation for bids, form a part of this specification.

SPECIFICATIONS

Military
JAN-P-125, Packaging and Packing for Overseas Shipment - Barrier-Materials, Waterproof, Flexible

DRAWINGS

Bureau of Ordnance
1402046 - Primer Container
1402047 - Fuse Prime Assembly
1402070 - Fuse Prime

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STANDARDS

Military
MIL-STD-129 - Marking of Shipments

American Society for Testing Materials
D541-49 - Standard Methods of Testing and Tolerances for Single Jute Yarn

(Copies of specifications, standards and drawings required by contractors in connection with specific procurement functions should be obtained from the procuring agency or as directed by the contracting officer.)

2.2 Other publications. - The following publication, of the issue in effect on date of invitation for bids, unless otherwise stated, forms a part of this specification.

Interstate Commerce Commission
49 CFR 71-78 - Interstate Commerce Commission Rules and Regulations for the Transportation of Explosives and Other Dangerous Articles.

(The Interstate Commerce Commission regulations are now a part of the Code of Federal Regulations (1949 Edition-Revised 1950) available from the Superintendent of Documents, Government Printing Office, Washington 25, D. C. Orders for the above publication should cite "49 CFR 71-78 (Rev. 1950).")

3. REQUIREMENTS

3.1 Construction. - The fuse shall consist of a continuous delay train of black powder having two cotton center threads in the train longitudinally. One center thread shall be 15/3 cotton, dyed to identify the burning time of the fuse (see paragraph 3.3). The second thread shall be 15/3 absorbent cotton and shall contain approximately 100% of its dry weight of water at the time of fuse core manufacture. The black powder train shall be covered with a tight spiral wrap of spinning yarns, Ten yarns shall be used wound in an S direction. Each yarn shall be 14 1/2 lb. jute spinning. Over the spinning yarns, ten yarns shall be tightly wound in the opposite direction (Z twist). One of these yarns shall be dyed to indicate the weight of powder per foot in the fuse (see paragraph 3.3). Each of these yarns shall be cotton designated 10/2. The textile core shall be covered with an extruded coating of polyethylene.

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3.2 Black powder. - The black powder used in the fuse shall be a potassium nitrate type containing 70 percent to 75 percent of potassium nitrate. Size classification of the black powder shall be at the option of the contractor and shall be so chosen and blended as to insure conformance of the fuse with the requirements of this specification.

3.3 Color of outer thread and center thread. - The color of the center and outer threads shall be in accordance with Table I.

TABLE I

Type	Color of Center Thread
I	Green
II	Yellow
III	Red

Class	Color of Thread in Outer Wrap
1	Green
2	Yellow

3.4 Cotton yarn. - The cotton yarn used in the manufacture of the fuse shall be natural, unfinished yarn spun from thoroughly clean, long staple carded cotton fiber. The cotton shall be evenly drawn, spun and twisted into a smooth and even yarn free from any lumps, lint or other imperfections which would adversely affect the finished fuse.

3.5 Jute yarn. - The jute yarn used in the manufacture of the fuse shall be made from pure jute fiber free from adulterants or foreign matter which will adversely affect the finished yarn. The yarn shall be untreated and of a natural color. The oil content of the yarn shall not be in excess of one percent and the volatile matter content shall not exceed 12.5 percent. All methods of testing and tolerances for the jute yarn shall conform to the requirements of ASTM Standard D541-49. All jute yarn shall be essentially level and free of bark, bunches, runrove, and large or loose knots.

3.6 Plastic covering. - The textile wrapping of the fuse shall be covered by a continuous waterproof coating of polyethylene having a minimum thickness of 0.012 inch. The polyethylene shall be of a grade equivalent to DYNF-3 polyethylene manufactured by the Bakelite Co., or Alathon-10 polyethylene manufactured by the E. I. du Pont Co.

3.7 Diameter. - The safety fuse shall have a nominal outside diameter of .200 inch. The fuse shall pass through a .215 inch ring gage and shall not pass through a .190 inch ring gage.

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3.8 Length. - The length of the fuse shall be as specified in the contract.

3.9 Flame intensity. - The fuse shall pass the flame intensity test designated in 4.4.2.

3.10 Burning time. - The burning time of the fuse shall be as shown in Table II.

TABLE II

Type	Time of Burning, Seconds per Foot*	
	Minimum	Maximum
I	27	33
II	40.5	49.5
III	54	66

*These burning times shall be obtained on fuse burned in the open at normal sea level barometric pressure.

3.11 Design characteristics. - The preproduction sample fuse shall meet the following design requirements in addition to those listed above.

3.11.1 Flame intensity at low and high temperatures. - The preproduction sample fuse shall pass the flame intensity test in 4.4.4.1 at +160°F ±5°F and -65°F ±5°F.

3.11.2 Burning time at low and high temperatures. - The burning time of the preproduction sample fuse at +160°F ±5°F and -65°F ±5°F shall be within the limits specified in 3.10 for the type of fuse tested.

3.11.3 Flexibility. - The preproduction sample fuse shall pass the flexibility test of 4.4.4.3.

3.11.4 Temperature and humidity. - The preproduction sample fuse after subjection to the temperature and humidity test of 4.4.4.4 shall meet the burning time requirements of 3.10.

3.11.5 Vibration. - The preproduction sample fuse after subjection to the 12 hour vibration test of 4.4.4.5 shall meet the burning time requirements of 3.10.

4. QUALITY ASSURANCE PROVISIONS

4.1 Sampling

4.1.1 Size of lot. - A lot shall not exceed 20,000 linear feet and shall consist of continuous production during any time period of 24 hours or less.

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4.1.2 Preproduction samples. - As soon as possible after award of the contract and prior to starting production the manufacturer shall fabricate 500 feet of preproduction sample fuse for shipment to a designated Government laboratory. Additional sample fuse shall be provided by the manufacturer on request by the Government Inspector in the case of changes in methods of fabrication, changes in materials and lengthy interruptions in production.

4.1.3 Production samples. - The Government Inspector shall select 10 percent of the reels or coils at random from each lot and cut from them a sufficient number of 18 inch fuse pieces for the tests. The group of sample fuse pieces shall be thoroughly mixed before testing.

4.2 Inspection. - Inspection shall be at the point of delivery unless otherwise specified in the contract order.

4.3 Classification of tests. - The inspection and testing of the plastic coated time fuse shall be classified as follows:

(a) Acceptance tests.
(b) Design tests.

4.4 Acceptance tests

4.4.1 Waterproofness test. - Seal each end of 20 of the 18 inch sample fuse pieces by crimping on an empty blasting cap shell with a double cut throat bench type cap crimper. Tie the two ends of the sample fuse pieces so that each piece forms a loop. Then suspend each loop in a vessel of water at 70° ±5°F to within 2 inches of the ends for 24 consecutive hours. Remove from the water, unfasten the ends, remove excess water, cut three inches off each end and ignite by means of a red hot iron or similar method. Measure the burning time of each fuse piece by the use of a stop watch or equivalent method which is capable of recording accurately to 0.1 second. The lot of fuse represented by the sample fuse pieces shall be rejected if one or more sample fuse pieces fail to burn completely at the burning rate specified in 3.10 for the type of fuse.

4.4.2 Flame intensity test. - Condition 20 of the 18 inch sample fuse pieces at 40 - 60 percent relative humidity and 70 ±5°F for not less than 24 consecutive hours. After conditioning, each fuse piece shall be immediately inserted and crimped in the fuse prime assembly as shown in Fig. 1. The opposite end of the fuse shall be ignited. The lot of fuses represented by the sample fuse pieces shall be rejected if two or more sample fuse pieces fail to ignite the fuse prime assembly crimped to each. In the case that one sample fuse piece fails to ignite the fuse prime assembly, the test shall be repeated with twice the number of samples. The lot shall be rejected if in the retest one or more fuse pieces fails to ignite the fuse

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prime assembly crimped to it. CAUTION: The operator should be protected from the ignition of the fuse prime assembly by a suitable shield or by maintaining sufficient distance from the flash.

4.4.3 Burning time test. - After conditioning 20 of the 18 inch sample fuse pieces at 40 - 60 percent relative humidity and 70 ±5°F for not less than 24 consecutive hours, ignite each fuse piece separately by means of a red hot iron or similar method. Measure the time of burning of each fuse piece by the use of a stop watch or equivalent method which is capable of recording accurately to 0.1 second. The lot of fuse represented by the fuse pieces shall be rejected if one or more sample fuse pieces fail to burn completely within the burning rate limits specified in 3.10, Table II, for the type of fuse under test.

4.4.4 Design tests. - The preproduction sample fuse shall pass the following design tests in addition to the acceptance tests listed above.

4.4.4.1 Flame intensity test at high and low temperatures. - Cut 20 fuse pieces 18 inches in length from the preproduction sample fuse. Condition the 20 sample fuse pieces at 40 - 60 percent relative humidity and +160°F ±5°F or -65°F ±5°F as applicable for not less than 24 hours. These fuse pieces shall be subjected to the flame intensity test as described in 4.4.2 while maintaining the same humidity and temperature under which the fuse pieces were conditioned. The preproduction sample shall not be approved if two or more fuse pieces fail to ignite the fuse prime assembly crimped to each. In the case one sample fuse piece fails to ignite the fuse prime assembly, the test shall be repeated with twice the number of samples. The preproduction sample shall not be approved if in the retest one or more fuse pieces fails to ignite the fuse prime assembly crimped to it. CAUTION: The operator should be protected from the ignition of the fuse prime assembly by a suitable shield or by maintaining sufficient distance from the flash.

4.4.4.2 Burning time test at high and low temperature. - Cut 20 fuse pieces 18 inches in length from the preproduction sample fuse. Condition the 20 fuse pieces at 40 - 60 percent relative humidity and +160°F ±5°F or -65°F ±5°F as applicable for not less than 24 consecutive hours. These fuse pieces shall be subjected to the burning time test of 4.4.3 while maintaining the same humidity and temperature under which the fuse pieces were conditioned. The preproduction sample shall not be approved if two or more fuse pieces fail to burn completely within the burning rate limits specified in 3.10, Table II, for the type of fuse under test. In case one fuse piece fails to meet the burning rate requirements, the test shall be repeated with twice the number of fuse pieces. The preproduction sample shall not be approved if in the retest one or more fuse pieces fail to burn completely within the specified burning rate limits for the type of fuse under test.

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4.4.4.3 Flexibility test. - Twenty fuse pieces approximately 10 inches long shall be cut from the preproduction sample fuse and subjected to the flexibility test. The plastic coating of the fuse piece shall not crack when the fuse piece is wrapped five times around a 0.50 inch diameter mandrel after being conditioned for 24 hours at -65°F ±5°F. This test shall be made with the fuze pieces and mandrel maintained at the conditioning temperature. The preproduction sample fuse shall not be approved if one or more fuse pieces show cracking of the plastic coating as a result of this test.

4.4.4.4 Temperature and humidity test. - Cut 20 fuse pieces 18 inches in length from the preproduction sample fuse. Seal each end of the 20 fuse pieces by crimping on an empty blasting cap shell with a double cut throat bench type cap crimper. Subject the sealed fuse pieces to the temperature and humidity test described in MIL-STD-354. After completion of the test remove excess moisture, cut three inches off each end and ignite by means of a red hot iron or similar method. Measure the burning time of each fuse piece by the use of a stop watch or equivalent method which is capable of recording accurately to 0.1 second. The preproduction sample shall not be approved if two or more fuse pieces fail to meet the burning rate requirements of 3.10 for the type of fuse under test. In case one sample fuse piece fails to meet the burning rate requirements, the test shall be repeated with twice the number of samples. The preproduction sample shall not be approved if in the retest one or more fuse pieces fail to meet the burning rate requirements of 3.10.

4.4.4.5 Vibration test. - Cut 20 fuse pieces 18 inches in length. Support the fuse pieces securely in a suitable test fixture and subject them to the vibration test described in MIL-STD-353. Following the vibration test cut three inches off each end of the fuse pieces and subject them to the burning time test of 4.4.3. The preproduction sample fuse shall not be accepted if two or more fuse pieces fail to burn completely within the burning rate limits specified in 3.10, Table II, for the type of fuse under test. In case one sample fuse piece fails to meet the burning rate requirements the test shall be repeated with twice the number of samples. The preproduction sample shall not be approved if in the retest one or more fuse pieces fail to burn completely within the burning rate limits specified in 3.10, Table II, for the type of fuse under test.

5. PREPARATION FOR DELIVERY

5.1 Packaging

5.1.1 Reels or coils. - The fuse shall be wound in coils of 50 feet each or on reels of 1,000 or 3,000 feet and wrapped with wrapping paper conforming to the requirements of JAN-P-125, Type E-1.

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5.2 Packing

5.2.1 Reels or coils. - Packaged reels or coils of fuse shall be packed in suitable waterproofed containers conforming with the requirements for shipment of safety fuse as specified in Interstate Commerce Commission's "Regulations for Transportation of Explosives and Other Dangerous Articles, Etc."

5.3 Marking

5.3.1 Reels or coils. - Each reel or coil of fuse shall be marked to show class, type of fuse, contractor's name, lot, inspector's initials, and date of manufacture.

5.3.2 Shipping containers. - Each inner and outer container shall be marked "TIME FUSE -- HANDLE WITH CARE", as specified by the rules and regulations of the Interstate Commerce Commission. In addition to any special marking required by the contract or order, all other marking shall conform to the requirements of MIL-STD-129.

6. NOTES

6.1 Data cards. - Place a data card 5 by 8 inches, made of commercial manila tag board (approximately 200 pounds ream weight) and properly filled out according to the form below, in each packing box. Two of these cards for each lot shall accompany each shipping ticket and six of these cards for each lot shall be furnished the Government Inspector.

AMMUNITION DATA CARD

KIND: Lot No. _____

Quantity in Lot _____

Spec. No. _____ Rev. _____

Dwg. No. _____ Rev. _____

Manufactured by _____

Contract No. _____

Packed _____

Remarks:

Certified to by _____

Inspector

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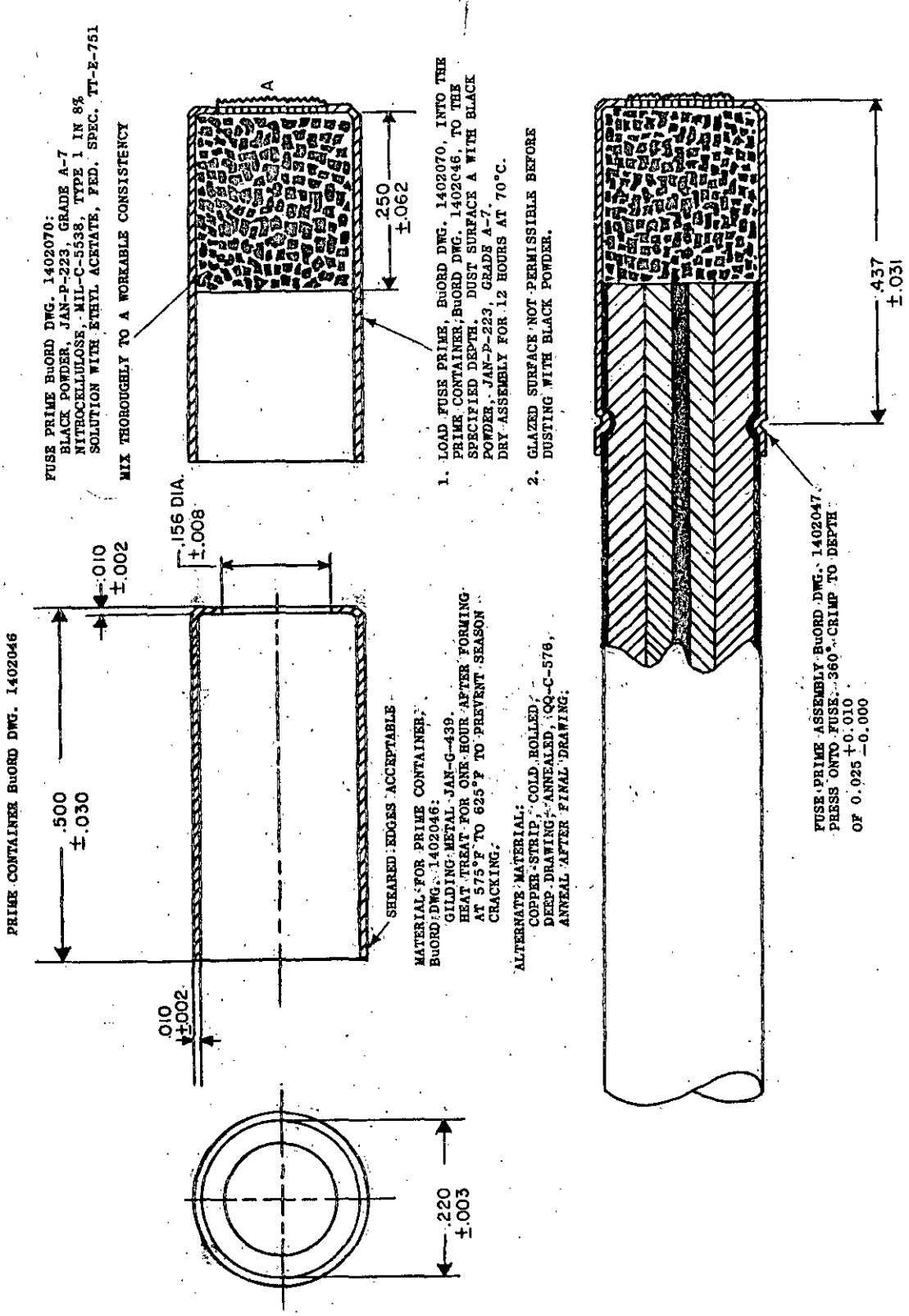


FIG. 1 FLAME INTENSITY TEST ASSEMBLY

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